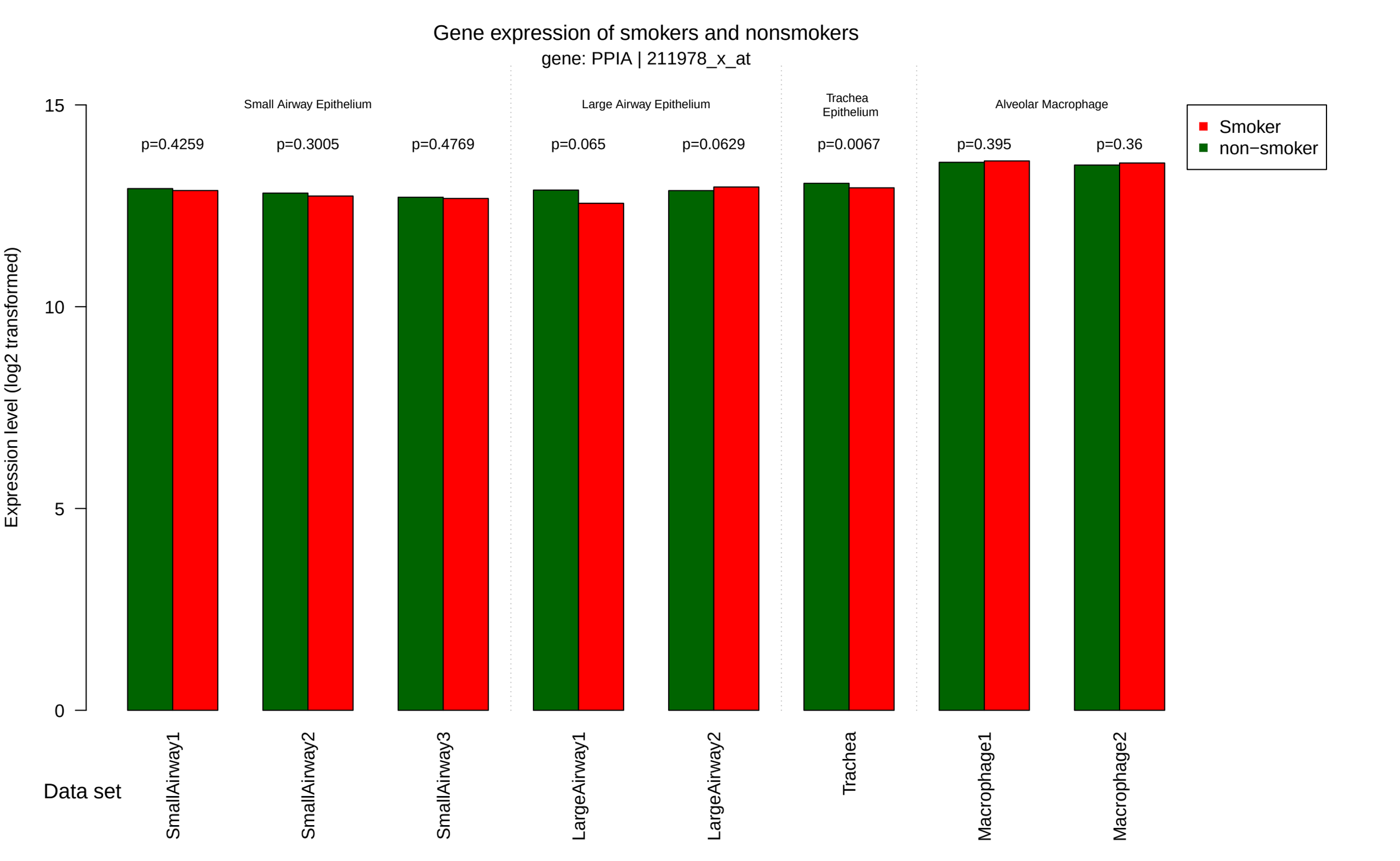
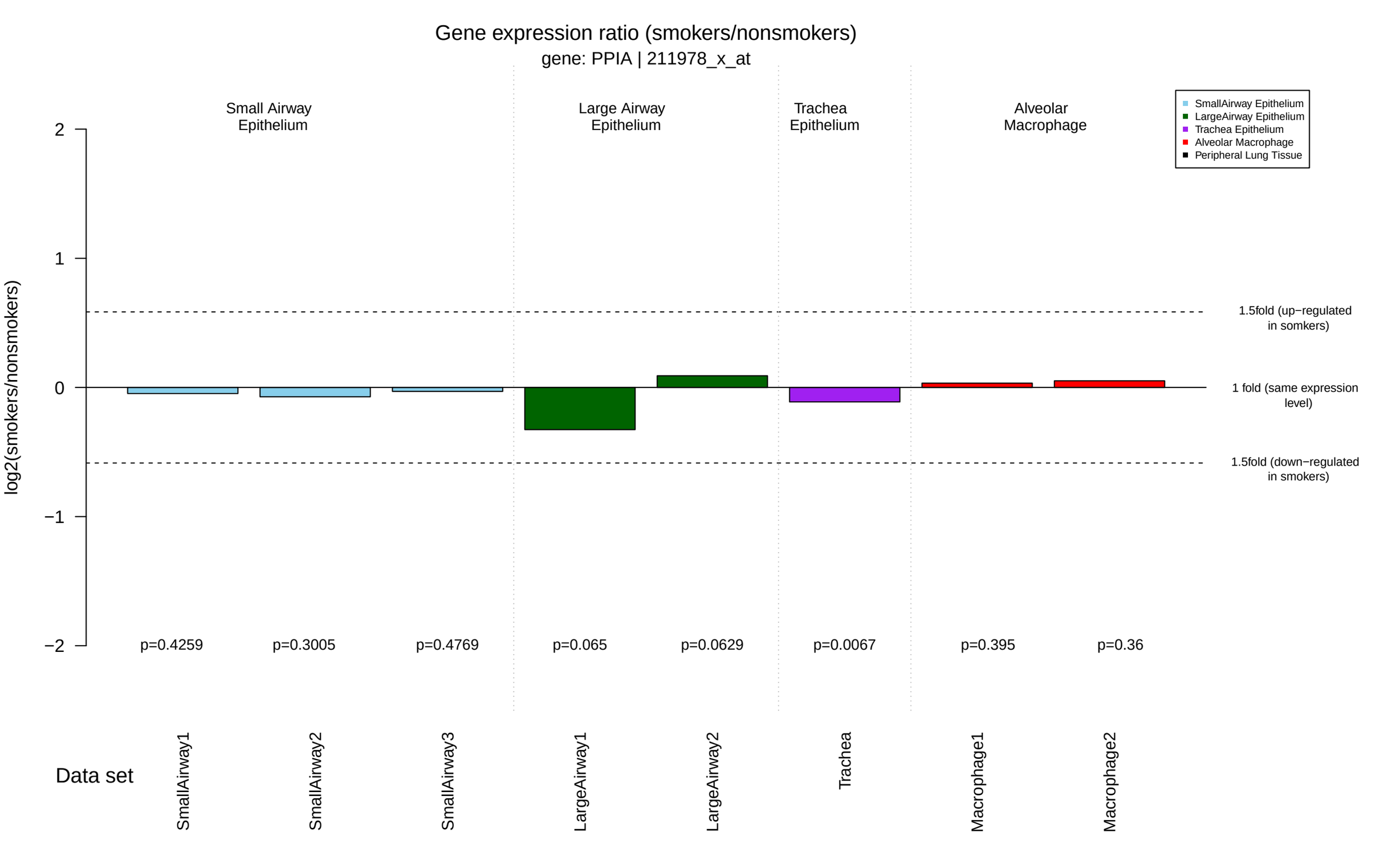
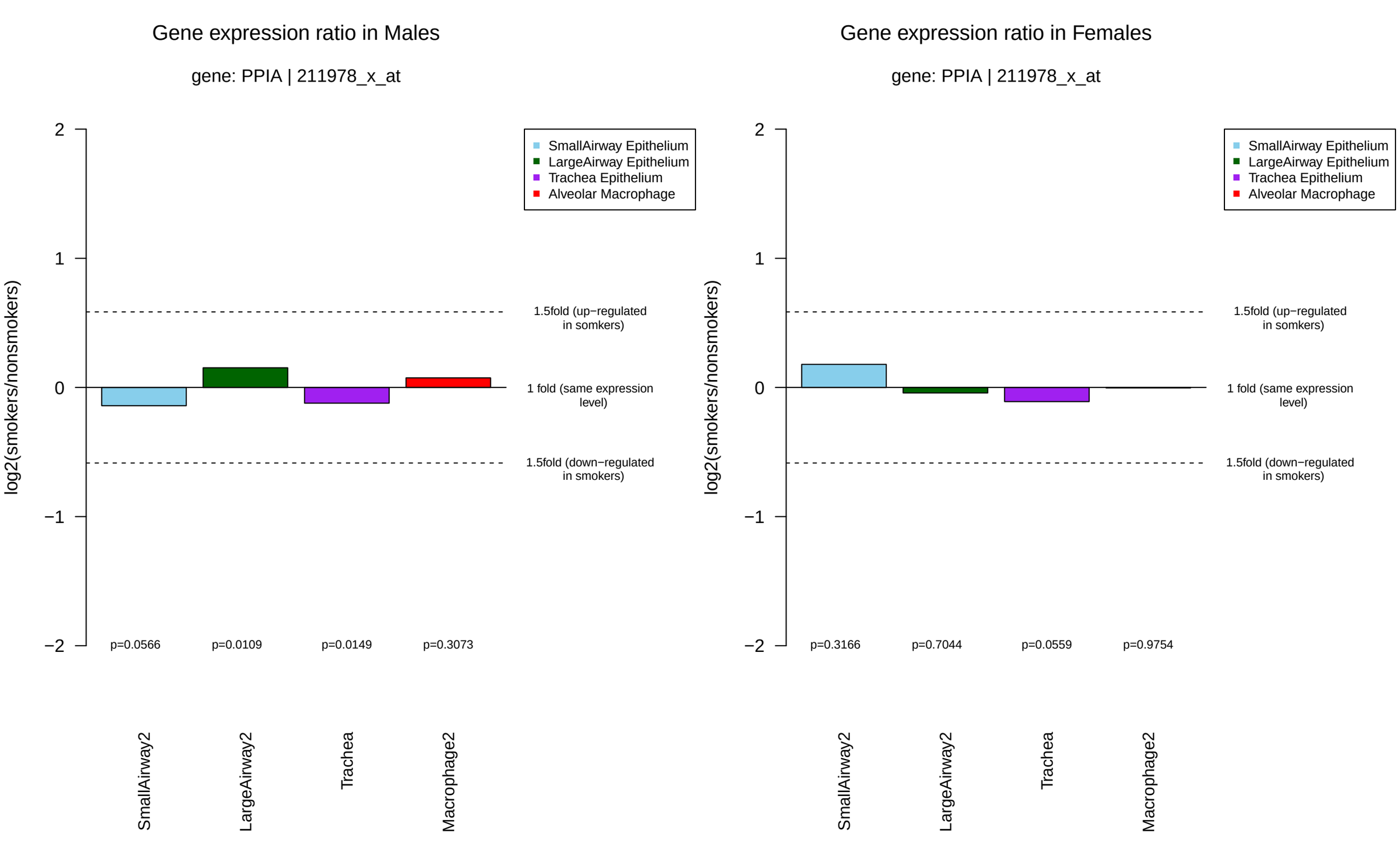
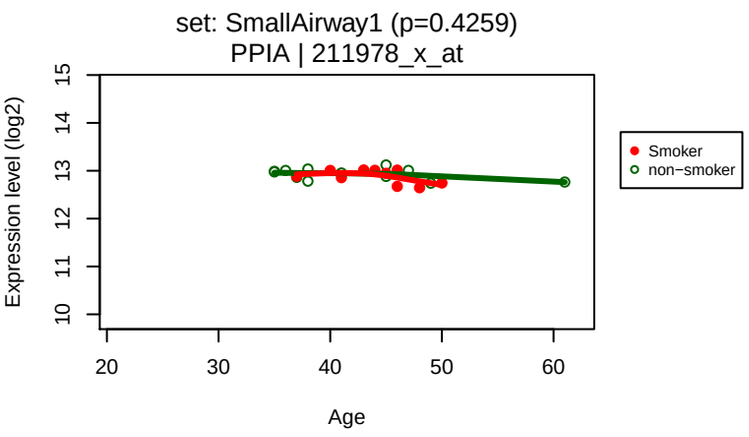
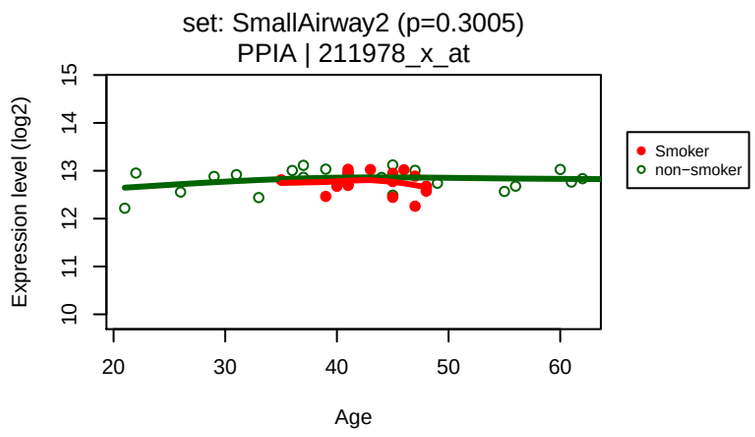
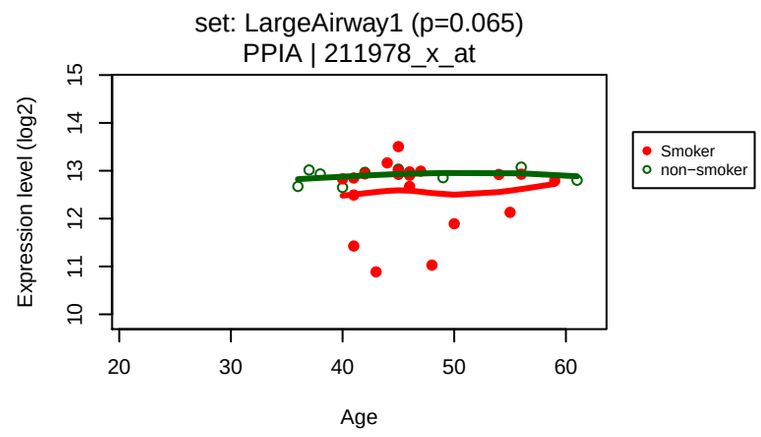

Gene expression of smokers and nonsmokers
gene: PPIA | 211978_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4259
p=0.3005
p=0.4769
p=0.065
p=0.0629
p=0.0067
p=0.395
p=0.36
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

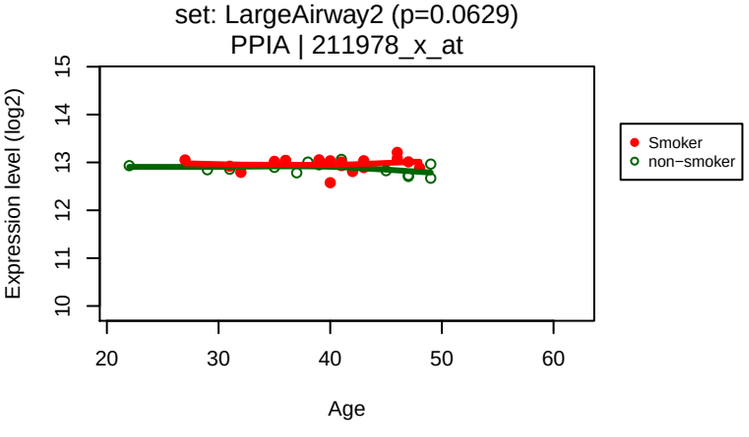

Gene expression ratio (smokers/nonsmokers)
gene: PPIA | 211978_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4259
p=0.3005
p=0.4769
p=0.065
p=0.0629
p=0.0067
p=0.395
p=0.36
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

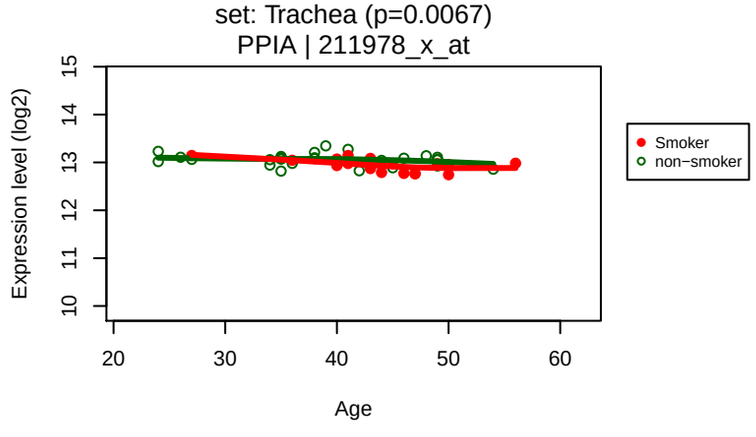

Gene expression ratio in Males
gene: PPIA | 211978_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0566
p=0.0109
p=0.0149
p=0.3073
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPIA | 211978_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3166
p=0.7044
p=0.0559
p=0.9754
SmallAirway2
LargeAirway2
Trachea
Macrophage2

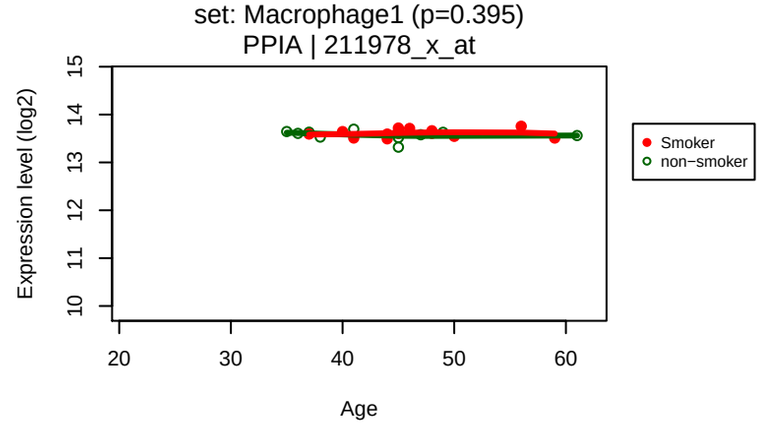

set: SmallAirway1 (p=0.4259)
PPIA | 211978_x_at
Expression level (log2)
Age
Smoker
non−smoker

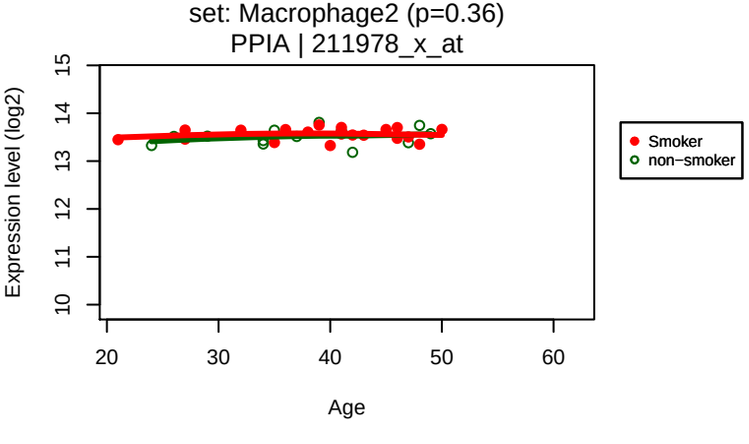

set: SmallAirway2 (p=0.3005)
PPIA | 211978_x_at
Expression level (log2)
Age
Smoker
non−smoker

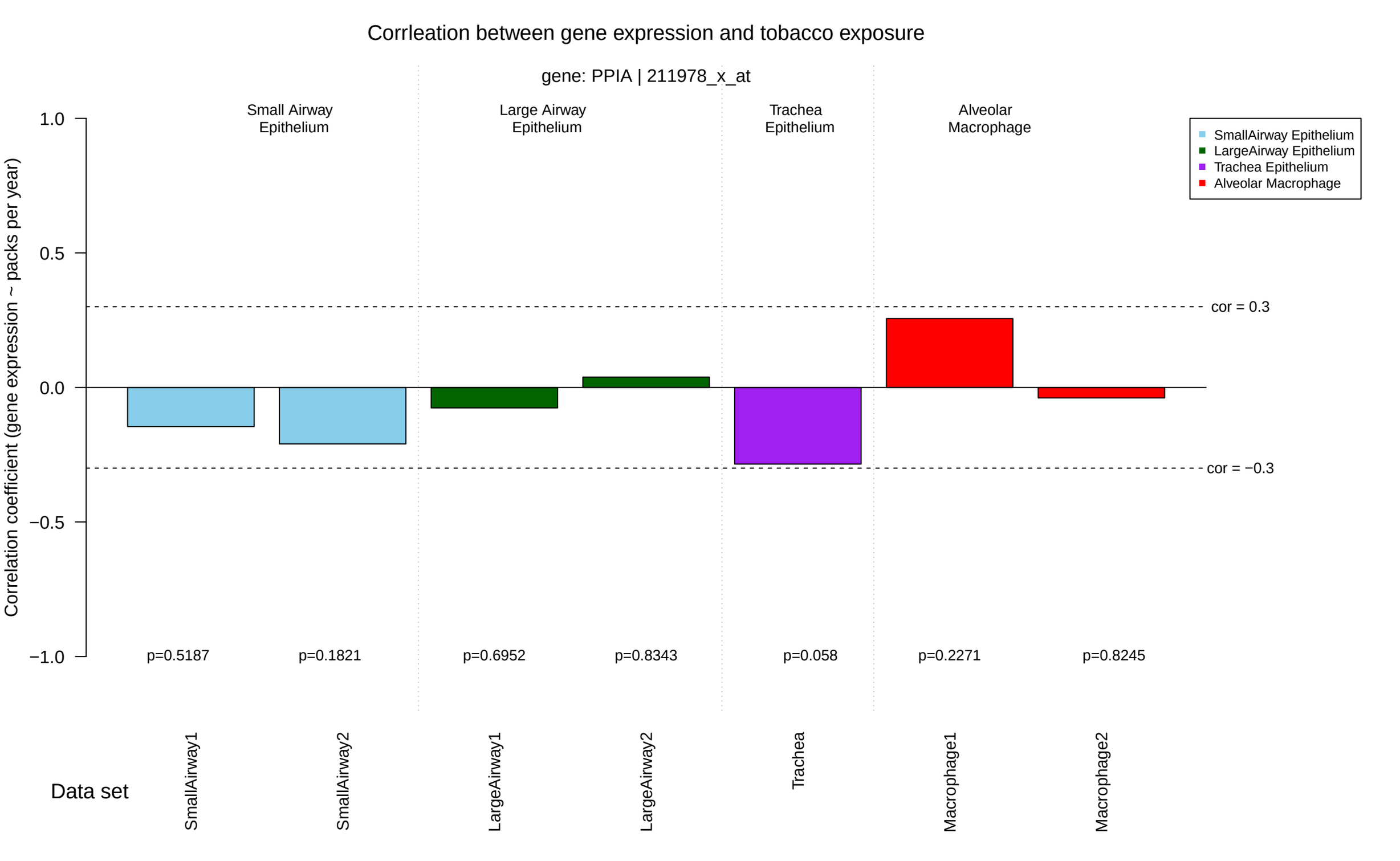

set: LargeAirway1 (p=0.065)
PPIA | 211978_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0629)
PPIA | 211978_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0067)
PPIA | 211978_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.395)
PPIA | 211978_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.36)
PPIA | 211978_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPIA | 211978_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5187
p=0.1821
p=0.6952
p=0.8343
p=0.058
p=0.2271
p=0.8245
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2